Trade the 8-K: Executive Exits Are Under-Priced, but Selling Puts Into Them Does Not Pay

Gator Quant Hacks 2026 · Massive bonus · Team: [team member names] · Code: [public GitHub URL], `strategy_8k/`

1. Hypothesis (committed before any option price was requested: `dbbfe6c`)

Categories: `ceo_departure` + `cfo_departure` + `executive_officer_departure`, pooled. **Strategy:** 5 · cash-secured put, 5% OTM, 3–6m expiry, entered at the close of the filing session. **Why the market should misprice it:** a leadership exit is vivid but ambiguous news. Nervous holders bid up downside puts (demand pressure, Gârleanu, Pedersen & Poteshman 2009), while for mega-caps the business rarely changes overnight. So the put seller earns an event premium on top of the ordinary variance risk premium, measured against a placebo of the same names on ordinary days. We pre-declared three predictions: (P1) CSP beats placebo at 21/42/exp; (P2) realized/implied move ratio is *below* placebo; (P3) the effect is stronger in 1m options. We only looked at event counts (189 in-sample) beforehand.

2. Method

The starter notebook's own cells run unchanged (`run_window.py` executes sections 1–7 for any `--start/--end`). Only the event table is replaced: the three tags are pooled, with one event per ticker within 5 days. Top-100 universe; spot comes from put-call parity; P&L is per $1 of spot; the placebo is 120 ordinary sessions for the same tickers at least 30 days from any event; 95% bootstrap CIs. In-sample: 2024-01-01..2025-12-31 (189 events, 77 tickers). Out-of-sample: 2026-01-01..2026-08-31, run once (67 events, 48 tickers).

3. Results at every fixed horizon (headline spec, events minus placebo, % of spot)

Sessions	1	2	3	5	10	21	42	63	exp
CSP, IS	−0.04	−0.13	−0.05	−0.24	−0.32	−0.38	+0.10	−0.06	+0.11
95% CI	±0.18	[−.37,.09]	±0.24	[−.58,.07]	[−.81,.14]	[−1.0,.25]	±0.98	±1.2	[−1.2,1.5]
CSP, OOS	+0.05	+0.20	+0.24	+0.21	+0.46	+0.66	**+1.62***	**+2.52***	+1.65
Stock only, OOS	+0.21	+0.72	+0.78	+0.79	+2.29	**+6.84***	**+8.97***	+8.43	+2.80

* = 95% CI excludes 0. OOS CIs: 42d [+0.14, +3.12], 63d [+0.76, +4.18]. In-sample, the best of the five strategies over placebo was the protective put (+0.25%), and none was significant at any horizon.

Verdict on the trade: it fails. In-sample, the CSP is indistinguishable from ordinary days at every horizon (P1 fails). Out-of-sample it turns positive, but the sign flip comes from the stock: exec-exit names rallied +6.8–9.0% versus ordinary days in 2026, and a short put is long delta. This is regime luck, not the volatility premium we hypothesized.

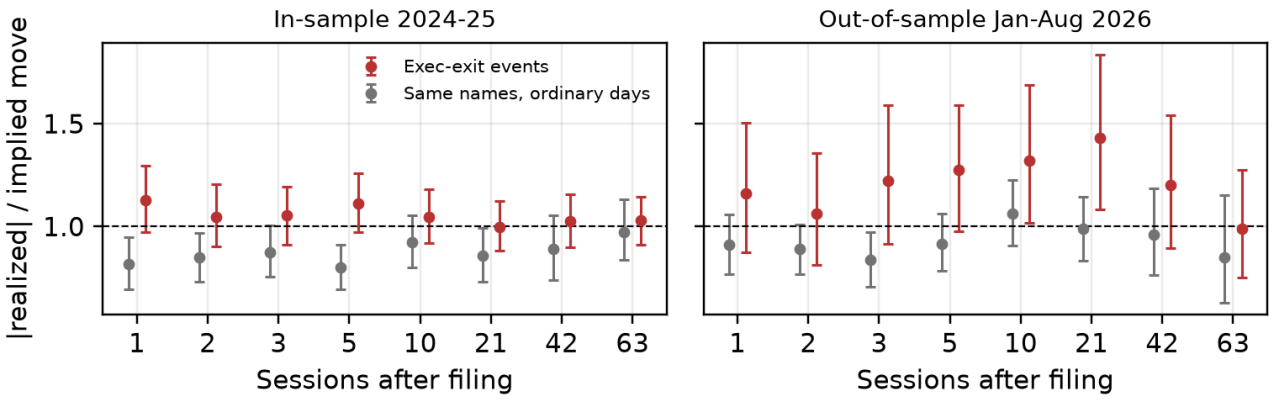


Figure. |Realized| ÷ implied move (3–6m ATM straddle read on the pre-event session, √t-scaled), mean and 95% CI. Red: exec-exit events; grey: same names on ordinary days.

What did replicate (the opposite of P2): after executive exits the stock moves **more** than the chain priced, relative to ordinary days. In-sample, 1–21 sessions: 1.00–1.13 vs placebo 0.80–0.92. Out-of-sample: 1.28–1.43 vs 0.92–1.06 at 5–21 sessions (21d: 1.43 [1.08, 1.83] vs 0.99 [0.83, 1.14]). The market treats an executive exit like an ordinary day and under-prices it. That is the opposite of the put-seller's story, and it explains why selling puts did not pay.

4. Sensitivity (pre-declared grid, `grid.py`)

We varied 3 expiry buckets × 2 entry times × 3 OTM levels (3/5/10%) × 2 category definitions (all exits vs CEO+CFO only) = 36 specs at 21/42/exp. Of the 54 tradeable (post-entry) cells, 5 are positive and **0** are significant. Mean CSP edge by bucket is 1m −0.54%, 2m −0.17%, 3–6m −0.10%: **P3 is reversed**. Short-dated puts sold after exits do worst, consistent with under-priced event risk. CEO+CFO-only (≈50 events) is negative in every cell.

5. Trials and other categories (all pre-registered in git, all reported)

- **Trial 2, post-hoc and labeled as such:** exec exits → protective put. OOS +6.1% (21d) and +7.7% (42d), but below the stock-only edge (+6.8% / +9.0%), so the option subtracted value. It fails the pre-registered credit rule.
- **Trial 3:** adverse 8-K at any ≥ $1B filer (cyber, disaster, closure, business exit, bankruptcy) → long call on top-100 peers sharing a 4-digit SIC code. IS 131 events: −0.6% / −1.4% at 21/42d; OOS 40 events: −1.0% / +2.1%, n.s. Fails. A naive SIC map paired micro-cap biotechs with MRK/LLY, so a point-in-time $1B materiality filter was required.
- **Trial 4:** `settlement_agreement` → CSP. IS 24 events, ≈0 then negative, and 1-day realized/implied 1.11 vs 0.85 (uncertainty did not fall). Only 5 OOS events: untestable.

6. What would break it, and how we would trade it

Break conditions: small samples (67 OOS events) and overlapping horizons; the static 2026 top-100 list (survivorship); filing date ≠ announcement time (most exits are pre-announced by press release; EDGAR acceptance timestamps would refine entry); daily last-trade marks instead of NBBO mid. **Trade (prediction for the sealed window):** the replicated finding points to a **delta-hedged long 1m straddle** on the filer, entered at the filing-session close and held 5–21 sessions. It is not tested here because it was not pre-registered. We predict a positive realized-minus-implied P&L on the judges' window, and we predict CSP on exec exits will *not* beat placebo there. **Realism:** top-100 ATM options trade nearly every session; at 1–2 contracts per event, ADV is not binding. With about 100 events a year this is a small-capacity, event-driven sleeve. **Sealed-window command:** `cd gqh-massive-8k-starter && .venv/bin/python ../strategy_8k/run_window.py --start A --end B --label sealed`.